MANU DWIVEDI

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SUMMARY

Director of Engineering, AI/ML R&D at Tiami Networks, leading a team of about 20 engineers. I build the machine learning that lets a telecom operator's existing radio network do more than carry traffic: sense drones, vehicles, and people as a radar; detect zero-day attacks on itself; and guide robots and drones that cannot navigate on their own. The portfolio runs on one shared stack: a radar foundation model, an RF digital twin for planning sensing coverage, and an edge deployment path from PyTorch research to licensed containers in operator networks.

LEADERSHIP

Tiami Networks

Director of Engineering, AI/ML R&D *Oct 2025 to Present*

- Lead about 20 engineers across machine learning, signal processing, RF, embedded systems, and robotics. Own the AI/ML roadmap, hiring, and delivery from research prototype to field deployment for carrier and defense customers.
- Set the product direction: integrated sensing and communications (ISAC) on infrastructure operators already own, so a sensing or security product needs no new spectrum and no new towers.
- Raised about **$6M** for sensing and ISAC research and productization. Sold counter-UAS sensors into approximately **every major US and EU telecom operator**. Brought PolyRAN to public trials with Vodafone and AirSpan.

PROGRAMS LED

PolyRAN: Radar Sensing on an Operator's Existing 5G Network **Tiami Networks**

2024 to Present

- Turns live 5G base stations into a wide-area radar without new radios, antennas, or changes to the RAN vendor's software. A sensing application (an O-RAN dApp) reads uplink reference signals the base station already schedules through a vendor-agnostic interface, and the radar foundation model turns them into detections and tracks.
- Runs on a server beside the base station's distributed unit; integrated with OpenAirInterface, OCUDU, and NVIDIA Aerial, with an optional fronthaul tap when the unit cannot export data.
- **Trialled publicly with Vodafone** at its Málaga R&D lab on live 5G, detecting objects and people without disrupting calls, messages, or data. Also trialled with **AirSpan** and tier-2 operators in Asia and the EU.

PolyEdge: Rapidly Deployable Passive Radar for Counter-UAS **Tiami Networks**

2023 to Present

- A standalone sensor for sites where the operator network cannot be modified. It listens to existing 5G downlink signals, so it needs no transmitter and no spectrum license; a software-defined receiver and edge compute run the same radar foundation model for detection, clutter rejection, and classification. Shipped as licensed containers (Docker, Helm, OpenShift).
- Detects small drones at about **1 km** (below 400 ft, up to 15 m/s, radar cross-section about 0.01 m^2) and identifies 5G-connected drones at 95% in clutter. A broadcast-TV variant (AirRadar) tracks light aircraft to about 25 km against ADS-B truth.

Agentic RF Planner: Digital Twins for ISAC **Tiami Networks**

2024 to Present

- Builds a city-scale RF digital twin from map and 3D building geometry. Vision-language agents read street-level imagery to assign building materials and clutter, and a physics-based propagation model plans communication coverage and radar sensing coverage on the same scene, showing where sensing will work before deployment. Used to site every Tiami deployment.
- Field-validated against commercial phone measurements in North America, Europe, and Asia across eight bands, standalone and non-standalone 5G. Representative case: **4.22 dB** mean absolute error over 17,324 measurements.

CoreDiff: Zero-Day Anomaly Detection for 5G SA and Private Networks **Tiami Networks; PIN-Cyber**

2025 to Present

- Converts any network into a structured graph of which components interact, and how, over time. A self-supervised model learns only normal behavior, and a latent diffusion model scores how unusual each interaction is, so new attack types are caught without signatures. Alarms are calibrated to a set false-alarm rate per hour, and each alert maps to MITRE ATT&CK.
- At the 1% anomalous tail: 5G SA about **94% F1**, with time to detect cut from 20 to 5 minutes. Private networks about **98.9% F1**, with mean time to detect cut from 120 to 0.6 seconds, across 13 ATT&CK techniques.

CoreDiffWFM: World Models That Turn ISAC into Autonomy Services **Tiami Networks**

2025 to Present

- Gives operators a service they can sell on top of their network: robots and drones that lack autonomy (no map, no global position, poor sensor coverage) receive it from the network. The world model fuses ISAC and PolyEdge detections, network positioning, fixed cameras, and each agent's own sensors into one shared picture, and returns goals, holds, and keep-out zones that the agent's own navigation executes. Agents' observations fill the picture where infrastructure coverage is poor.

- **Indoor:** a ground robot with no map or position is localized from fixed cameras and its lidar (offline runs), the room is rebuilt from a scan and camera views, and no goal is sent until position and keep-out zones exist. **Outdoor:** drone and ground-robot fleets over city geometry, where a hazard seen by one agent replans every affected mission.
- The RF planner and a ray-traced physics model predict radio conditions along each route at every time step. This explains how the network will behave for an agent that must keep its link through a mission, and picks routes and upload points that finish sooner on the existing network.
- A predicted 1.000 m move was observed as 0.980 m on a robot testbed (within three-sigma uncertainty). In simulation, three drones avoided a helicopter and cut fleet completion from 152 to 125 seconds; ground robots finished 6 of 6 missions around a blockage versus 3 of 6.

ARCHES: End-to-End Counter-UAS Response **Tiami Networks**
2025 to Present

- Closes the loop from detection to response. A radar cue hands off to an interceptor drone that tracks the target with onboard vision (YOLOv8) and Remote-ID, working with commercial drones without a vendor SDK. A staged safety process takes flight control from simulation to real flight on NVIDIA Jetson, with a ground station coordinating several drones.

RF Pattern-of-Life **Tiami Networks**
2025 to Present

- Identifies and locates radio emitters (Wi-Fi, drone control links, 5G) from raw RF without decoding them, using channelization, burst fingerprinting, and four-antenna angle of arrival. Won over **$1M** in follow-on R&D funding.

EARLIER EXPERIENCE

Tiami Networks
Principal Embedded ML Engineer *Dec 2024 to Oct 2025*

- Led sensing proofs of concept with Tier-1 mobile operators, MVNOs, a physical security firm, and the US Army. Uplink 5G pedestrian tracking with NVIDIA and Northeastern University, and ISAC fall detection with Deutsche Telekom (MWC 2025). Pedestrian and traffic monitoring for the City of Elk Grove at 99.6% accuracy from 250 m to 1.3 km.

Tiami Networks
Sr. Embedded ML Engineer and AI FPGA Intern *Sept 2023 to Dec 2024*

- Took the first 5G sensing models from software-radio prototypes to production on Intel and FPGA hardware. Closed-loop network assurance with 3GPP NWDAF won the Deutsche Telekom T-Challenge 2024 Most Integration Potential Award (EUR 25,000).

RadicalX
AI Engineer Intern *Jul 2023 to Sept 2023*

- Exam anti-cheating vision model at **96%** accuracy (attention-based CNN), deployed through AWS SageMaker CI/CD; GPT-3 fine-tuning in Docker on GPU clusters.

Michigan State University
Research Assistant *May 2022 to Jun 2023*

- Generative-AI acceleration on low-power FPGAs with a **45% compute reduction** without pruning or quantization. Graduate Office Fellowship ($7,500).

OpenPower Foundation at IBM
Researcher *Aug 2020 to Dec 2021*

- Open-ISA processor accelerators for data access and matrix multiply; about 4x ALU throughput for GenAI workloads.

PUBLICATIONS AND PATENTS

- M. Dwivedi et al., "5G-Based Passive Radar Sensing for Human Activity Recognition using Deep Learning," *IEEE RadarConf*, 2024.
- M. Dwivedi et al., "Person Presence Detection using 5G Synchronization Signals on SDR Testbed," *IEEE MILCOM* (demo), 2024.
- Two U.S. patents pending: indoor and outdoor RF activity sensing; cross-modal radar and camera sensing.

TECHNICAL SKILLS

ML : PyTorch, TensorFlow, self-supervised and diffusion models, vision-language agents, YOLO, Gaussian splatting, world models

Deployment : CUDA, TensorRT, ONNX, NVIDIA Jetson, FPGA (Vitis AI), Docker, Kubernetes/Helm, OpenShift, AWS

Telecom : 5G NR, O-RAN dApps and E3, OpenAirInterface, NVIDIA Aerial, passive radar, Sionna RT, 3GPP TR 38.901

Robotics : ROS 2, Nav2, PX4, Gazebo; Python, C/C++, CUDA C

EDUCATION

- **Michigan State University**, M.S. Electrical and Computer Engineering, GPA 3.8
- **SRM Institute of Science and Technology**, B.Tech. Electronics and Communication Engineering, GPA 3.8